

Q.1

$$K = \begin{bmatrix} 53.28 & 0 & -266.4 & 53.28 & 0 & -266.4 & 0 & 0 & 0 \\ 0 & 4800 & 0 & 0 & -24800 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 888 & 0 & 0 & 0 \\ -266.4 & 0 & 1776 & 266.4 & 0 & 888 & 0 & 0 & 0 \\ -53.28 & 0 & 266.4 & 118553 & 0 & 266.4 & -118500 & 0 & 0 \\ 0 & -24800 & 0 & 0 & 24004 & 416.3 & 0 & -104 & 416.3 \\ -266.4 & 0 & 888 & 266.4 & 416.5 & 3006 & 0 & -416.3 & 1110 \\ 0 & 0 & 0 & -118500 & 0 & 0 & -118500 & 0 & 0 \\ 0 & 0 & 0 & 0 & -104 & -416.3 & 0 & 104 & -416.3 \\ 0 & 0 & 0 & 0 & 416.3 & 1110 & 0 & -416.3 & 2220 \end{bmatrix}$$

$$U = \begin{bmatrix} 0 \\ 0 \\ 0.01 \\ 4.2 \\ 2.2 \\ 0.2 \\ 0.05 \\ 0.10 \\ 0.3 \end{bmatrix}$$

$$F = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$F^F = \begin{bmatrix} -120 \\ 0 \\ 200 \\ -120 \\ 37.5 \\ -125 \\ 0 \\ 37.5 \\ -75 \end{bmatrix}$$

$$U_f = \begin{bmatrix} U_2 \\ v_2 \\ \theta_2 \\ \theta_3 \end{bmatrix}$$

$$U_{re} = \begin{bmatrix} 0 \\ 0 \\ 0.01 \\ 0.005 \\ 0.1 \end{bmatrix}$$

$$K_{ff} U_f = F - F^F - K_{re} U_{re}$$

$$\begin{bmatrix} 53.28 & 118553 & 0 & 266.4 & 0 & 416.25 \\ 0 & 0 & 24004 & 416.25 & 416.25 & 1110 \\ 0 & 0 & 416.25 & 3006 & 1110 & 2220 \\ -266.4 & 266.4 & 416.25 & 1110 & 2220 & 2220 \end{bmatrix} \begin{bmatrix} U_2 \\ v_2 \\ \theta_2 \\ \theta_3 \end{bmatrix} = \begin{bmatrix} -120 \\ 37.5 \\ -12.5 \\ -75 \end{bmatrix}$$

$$- \begin{bmatrix} 53.28 & 0 & -266.4 & 0 & 0 & 0 \\ 0 & 4800 & 0 & 0 & 0 & 0 \\ -266.4 & 0 & 1776 & 0 & 0 & 0 \\ 0 & 0 & 0 & -118500 & 0 & 0 \\ 0 & 0 & 0 & 0 & 104 & -416.3 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0.01 \\ 0.005 \\ 0.1 \end{bmatrix}$$

$$- \begin{bmatrix} 53.28 & 0 & +266.4 & -118500 & 0 \\ 0 & -24900 & 0 & 0 & -104 \\ -266.4 & 0 & 888 & 0 & -416.25 \\ 0 & 0 & 0 & 0 & -416.25 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0.01 \\ 0.005 \\ 0.01 \end{bmatrix}$$

$$K_{free} U_{re} = \begin{bmatrix} -580.8 \\ 1.04 \\ 13.04 \\ 4.16 \end{bmatrix} \quad \therefore K_{ff} U_f = \begin{bmatrix} -120 \\ 37.5 \\ -125 \\ -75 \end{bmatrix} - \begin{bmatrix} -580.8 \\ 1.04 \\ 13.04 \\ 4.16 \end{bmatrix}$$

$$\therefore K_{ff} U_f = \begin{bmatrix} -700.8 \\ -38.5 \\ 111.06 \\ 70.8 \end{bmatrix}$$

$$\therefore U_f = \begin{bmatrix} 0.00594 \\ -0.000594 \\ 0.02182 \\ 0.02111 \end{bmatrix}$$

$$U = \begin{bmatrix} 0 \\ 0 \\ 0.01 \\ 0.00594 \\ -0.000594 \\ 0.02182 \\ 0.005 \\ -0.01 \\ 0.02111 \end{bmatrix}$$

Reaction force:

$$R = K_u + f_f$$

$$= K \begin{bmatrix} 0 \\ 0 \\ 0.01 \\ 0.00594 \\ 0.00594 \\ 0.02182 \\ 0.005 \\ -0.01 \\ 0.02111 \end{bmatrix} + \begin{bmatrix} -120 \\ 0 \\ 200 \\ -120 \\ 37.5 \\ -125 \\ 0 \\ 37.5 \\ -75 \end{bmatrix} = \begin{bmatrix} -8.8 \\ 56.3 \\ 38.72 \\ 120 \\ -37.5 \\ 125 \\ -111.2 \\ -18.8 \\ 75 \end{bmatrix} + \begin{bmatrix} -120 \\ 0 \\ 200 \\ -120 \\ 37.5 \\ -125 \\ 0 \\ 37.5 \\ -75 \end{bmatrix} = \begin{bmatrix} -128.8 \\ 56.3 \\ 238.7 \\ 0 \\ 0 \\ 0 \\ -111.2 \\ 18.65 \\ 0 \end{bmatrix}$$

$$\therefore R_{1x} = -128.8 \text{ kN}$$

$$R_{2y} = 56.3 \text{ kN}$$

$$M_1 = 238.7 \text{ kN}$$

$$R_{3x} = -111.2 \text{ kN}$$

$$B_{3y} = 18.65 \text{ kN}$$

Global equilibrium:

$$\text{Ext force (x dir)} = 24 \times 10 = 240 \text{ kN}$$

$$\text{" " (y dir)} = 75 \text{ kN}$$

$$\sum F_x = 240 - 128.8 - 111.2 = 0$$

$$\sum F_y = 56.3 + 18.65 - 75 = 0$$

(OK)

Local element end ~~force~~ displacement:

$$U_1 = T U_G$$

$$U_1 = T U_G$$

$$= \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ .01 \\ .00594 \\ .00594 \\ .02182 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ .01 \\ -.00594 \\ -.00594 \\ .02182 \end{bmatrix}$$

$$U_2 = I U_{G(2)}$$

$$= \begin{bmatrix} .00594 \\ -.00594 \\ .02182 \\ .005 \\ -.01 \\ .02111 \end{bmatrix}$$

Local element end force:

$$Q_{1(1)} = K_{11} U_{1(1)} + Q_{1(1)}^F$$

$$= \begin{bmatrix} 24800 & 0 & 0 & -24800 & 200 & 0 \\ 0 & 53.28 & 266.4 & 0 & -53.28 & 266.4 \\ 0 & 266.4 & 1776 & 0 & -266.4 & 888 \\ -24800 & 0 & 0 & 24800 & 0 & 0 \\ 0 & -53.28 & -266.4 & 0 & 53.28 & -266.4 \\ 0 & 266.4 & 888 & 0 & -266.4 & 1776 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ .01 \\ -.00594 \\ -.00594 \\ .02182 \end{bmatrix}$$

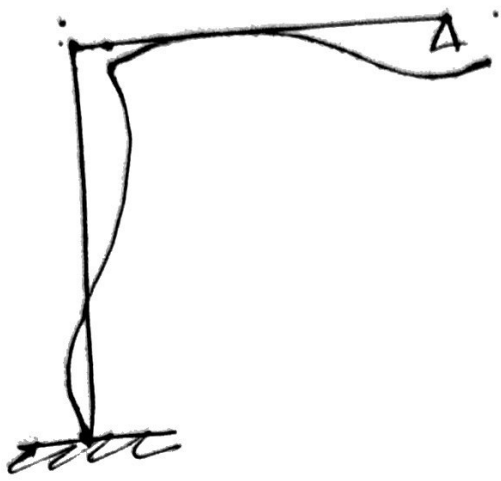
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$$+ \begin{bmatrix} 0 \\ 120 \\ 200 \\ 0 \\ 120 \\ -200 \end{bmatrix} = \begin{bmatrix} 56.35 \\ 8.79 \\ 38.72 \\ -56.55 \\ -8.8 \\ 49.21 \end{bmatrix} + \begin{bmatrix} 0 \\ 120 \\ +200 \\ 0 \\ +120 \\ -200 \end{bmatrix} = \begin{bmatrix} 56.35 \\ 128.793 \\ 238.78 \\ -56.348 \\ 111.207 \\ -150.787 \end{bmatrix}$$

$$q_{x(2)} = K_2 U_{x(2)} + Q_2 F$$

$$= K_2 \begin{bmatrix} .00594 \\ -.000594 \\ .02182 \\ .005 \\ -.01 \\ .0211 \end{bmatrix} + \begin{bmatrix} 0 \\ 37.5 \\ 75 \\ 0 \\ 37.5 \\ -75 \end{bmatrix} = \begin{bmatrix} 111.207 \\ 56.348 \\ 150.787 \\ -111.207 \\ 18.652 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 111.207 \\ 18.85 \\ 75.70 \\ -111.21 \\ -18.85 \\ 75 \end{bmatrix} + \begin{bmatrix} 0 \\ 37.5 \\ 75 \\ 0 \\ 37.5 \\ -75 \end{bmatrix} = \begin{bmatrix} 111.207 \\ 56.348 \\ 150.787 \\ -111.207 \\ 18.652 \\ 0 \end{bmatrix}$$



For Q2,
Uniform temp change for element 1

$$\Delta = +10^\circ\text{C}$$

$$\alpha = 12 \times 10^{-6}$$

$$L = 10$$

$$\begin{aligned} \therefore EA\alpha\Delta &= 200 \times 10^9 \times 4740 \times 10^{-6} \times 12 \times 10^{-6} \times 10 \\ &= 0.48000 \times 12 \times 10^5 \\ &= 113.76 \text{ kN} \end{aligned}$$

$$q_1^H = EA\alpha T \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 113.76 \\ 0 \\ 0 \\ -113.76 \\ 0 \\ 0 \end{bmatrix}$$

for element 2,

$$d = 0.3 \text{ m}$$

$$T_t = 15^\circ\text{C}$$

$$T_b = 5^\circ\text{C}$$

$$\therefore \frac{A(T_b + T_t)}{2} = \frac{4740 \times 10^{-6} \times (15 + 5)}{2}$$

$$= 474 \times 10^{-3}$$

$$\frac{I(T_b - T_t)}{d} = \frac{22.2 \times 10^6 \times 10^{-12} \times (5 - 15)}{0.03}$$

$$= -0.637$$

$$\begin{aligned} E\alpha &= 200 \times 10^9 \times 12 \times 10^{-6} \\ &= 2.4 \times 10^6 \end{aligned}$$

$$\therefore E\alpha \frac{A(T_b + T_t)}{2} = 113.76 \text{ kN}$$

$$\frac{E\alpha I(T_b - T_t)}{d} = -1.776 \text{ kN}$$

$$\therefore q_2^H = E\alpha \begin{bmatrix} A(T_b + T_t)/2 \\ 0 \\ I(T_b - T_t)/d \\ -A(T_b + T_t)/2 \\ 0 \\ -I(T_b - T_t)/d \end{bmatrix} = \begin{bmatrix} 113.76 \\ 0 \\ -1.776 \\ -113.76 \\ 0 \\ 1.776 \end{bmatrix}$$

Element 1:

$$Q_1 = T_1^T q^H,$$

$$= \begin{bmatrix} 0 & 1 & & & & \\ & -1 & & & & \\ & & 0 & 0 & 1 & \\ & & 0 & 0 & 0 & 0 & 1 & 0 \\ & & 0 & 0 & 0 & -1 & & \\ & & & & & & & 1 \end{bmatrix} \begin{bmatrix} 113.76 \\ 0 \\ 0 \\ -113.76 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ +113.76 \\ 0 \\ 0 \\ -113.76 \\ 0 \end{bmatrix}$$

$$Q_2 = T_2^T q^H$$

$$= I q_2^H$$

$$= q_2^H$$

$$\therefore Q = \begin{bmatrix} 0 \\ +113.76 \\ 0 \\ 0 + 113.76 \\ -113.76 + 0 \\ 0 + -1.776 \\ -113.76 \\ 0 \\ 1.776 \end{bmatrix} = \begin{bmatrix} 0 \\ +113.76 \\ 0 \\ +113.76 \\ -113.76 \\ -1.776 \\ -113.76 \\ 0 \\ 1.776 \end{bmatrix}$$

$$\therefore F^F + F^H = \begin{bmatrix} -120 \\ 0 \\ 200 \\ -120 \\ 37.5 \\ -125 \\ 0 \\ 37.5 \\ -75 \end{bmatrix} + \begin{bmatrix} 0 \\ +113.76 \\ 0 \\ +113.76 \\ -113.76 \\ -1.776 \\ -113.76 \\ 0 \\ 1.776 \end{bmatrix} = \begin{bmatrix} -120 \\ +113.76 \\ 200 \\ -6.24 \\ -76.26 \\ -126.776 \\ -113.76 \\ 37.5 \\ -73.224 \end{bmatrix}$$

Free DOF = 4, 5, 6, 9

$$F = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\therefore R.H.S = F - F^F - F^H = \begin{bmatrix} 6.24 \\ 76.26 \\ 126.776 \\ 73.224 \end{bmatrix}$$

$$K_f U = F - F^H - F^H$$

solving $U_f = \begin{bmatrix} -0.000006165 \\ +0.000602 \\ 0.2617 \\ 0.01979 \end{bmatrix}$

$$\therefore U = \begin{bmatrix} 0 \\ 0 \\ 0 \\ -0.000006165 \\ +0.000602 \\ 0.2617 \\ 0 \\ 0 \\ 0.01979 \end{bmatrix}$$

Reactions:

$$R = KU + F_f + F^H - F$$

$$= K \begin{bmatrix} 0 \\ 0 \\ 0 \\ -0.000006165 \\ +0.000602 \\ 0.2617 \\ 0 \\ 0 \\ 0.01979 \end{bmatrix} + \begin{bmatrix} -120 \\ +113.76 \\ 200 \\ -6.24 \\ -76.26 \\ -126.776 \\ -113.76 \\ 37.5 \\ -73.224 \end{bmatrix} = \begin{bmatrix} -60.71 \\ -57.069 \\ 23.235 \\ 6.24 \\ 76.26 \\ 126.776 \\ 73.05 \\ -19.19 \\ 73.224 \end{bmatrix} + \begin{bmatrix} -120 \\ 113.76 \\ 200 \\ -6.24 \\ -76.26 \\ -126.776 \\ -113.76 \\ 37.5 \\ -73.224 \end{bmatrix}$$

$$= \begin{bmatrix} -126.97 \\ 56.69 \\ 223.23 \\ 0 \\ 0 \\ 0 \\ -113.03 \\ 18.309 \\ 0 \end{bmatrix}$$

$$\begin{aligned} R_{1x} &= -126.97 \text{ kN} \\ R_{1y} &= 56.69 \text{ kN} \\ M_{1x} &= 223.23 \text{ M} \\ R_{3x} &= -113.03 \text{ kN} \\ R_{3y} &= 18.309 \text{ kN} \end{aligned}$$

Global eq check:

$$\begin{aligned} \sum F_x &= -126.97 - 113.03 + 24 \times 10 = 0 \\ \sum F_y &= -75 + 56.69 + 18.31 = 0 \quad (\text{ok}) \end{aligned}$$

Local element disp:

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$$u_1^1 = T_1 U_1^G$$

$$= \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ -800006165 \\ 000602 \\ 02617 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 000602 \\ 000006165 \\ 02617 \\ 0 \end{bmatrix}$$

$$u_2^2 = T_2 U_2^G = I U_2^G = \begin{bmatrix} 000006165 \\ 000602 \\ 02617 \\ 0 \\ 01070 \\ 0 \end{bmatrix}$$

Local element end force:

$$q_1^1 = k_1^1 u_1^1 + Q_1^{Tot(X)}$$

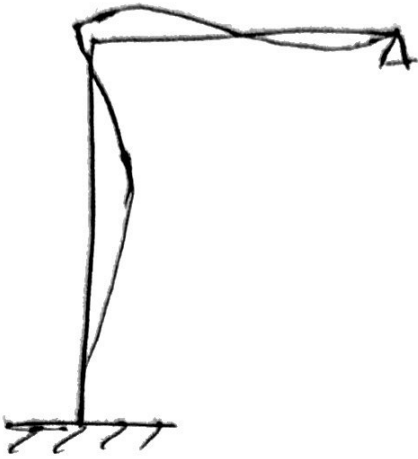
$$= k_1^1 \begin{bmatrix} 0 \\ 0 \\ 000602 \\ 000006165 \\ 02617 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 120 \\ 200 \\ 0 \\ 120 \\ -200 \end{bmatrix} + \begin{bmatrix} 113.76 \\ 0 \\ -113.76 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} -57.069 \\ 6.97 \\ 23.235 \\ 57.069 \\ -6.971 \\ 46.471 \end{bmatrix} + \begin{bmatrix} 0 \\ 120 \\ 200 \\ 0 \\ 120 \\ -200 \end{bmatrix} + \begin{bmatrix} 113.76 \\ 0 \\ -113.76 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 56.691 \\ 126.071 \\ 223.235 \\ -56.691 \\ 113.029 \\ -153.529 \end{bmatrix}$$

$$q_2^1 = k_2^1 u_2^1 + Q_2^{Tot}$$

$$= k_2^1 \begin{bmatrix} 000006165 \\ 000602 \\ 02617 \\ 0 \\ 0 \\ 01070 \end{bmatrix} + Q_2^{Hx} + Q_2^{F1}$$

$$= \begin{bmatrix} -7305 \\ 101011 \\ 80305 \\ 0735 \\ -10101 \\ 73224 \end{bmatrix} + \begin{bmatrix} 113.76 \\ 0 \\ -1776 \\ -113.76 \\ 0 \\ 1776 \end{bmatrix} + \begin{bmatrix} 0 \\ 37.5 \\ 75 \\ 0 \\ 37.5 \\ -75 \end{bmatrix} = \begin{bmatrix} 113.03 \\ 56.69 \\ 153.53 \\ -113.03 \\ 18.009 \\ 0 \end{bmatrix}$$



Q.3

Both elements are 0.003 m short.

$$\frac{EA}{L} e_a = \frac{200 \times 10^9 \times 4740 \times 10^{-6} \times 0.003}{L}$$

$$= \frac{-2844}{L} \text{ kN}$$

for element 1,

$$\frac{EA}{L_1} e_a = \frac{-2844}{10} \text{ kN} = -284.4 \text{ kN}$$

for element 2,

$$\frac{EA}{L_2} e_a = \frac{-2844}{8} \text{ kN} = -355.5 \text{ kN}$$

$$\therefore q_1^E = \frac{EA}{L} e_a \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -284.4 \\ 0 \\ 0 \\ 284.4 \\ 0 \\ 0 \end{bmatrix}$$

$$q_2^E = \begin{bmatrix} -355.5 \\ 0 \\ 0 \\ 355.5 \\ 0 \\ 0 \end{bmatrix}$$

$$q_1^G = T_1^T q_1^E = \begin{bmatrix} 0 & -1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -284.4 \\ 0 \\ 0 \\ 284.4 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -284.4 \\ 0 \\ 0 \\ 284.4 \\ 0 \end{bmatrix}$$

$$q_2^G = T_2^T q_2^E = I q_2^E = q_2^E$$

$$\therefore F^E = \begin{bmatrix} 0 \\ -284.4 \\ 0 \\ -355.5 \\ 284.4 \\ 0 \\ 355.5 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$F_{tot} = \begin{bmatrix} 0 \\ -120 \\ 0 \\ 200 \\ -120 \\ 37.5 \\ -12.5 \\ 0 \\ 37.5 \\ -7.5 \end{bmatrix} + \begin{bmatrix} 0 \\ -284.4 \\ 0 \\ -355.5 \\ 284.4 \\ 0 \\ 355.5 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -120 \\ -284.4 \\ 200 \\ -475.5 \\ 321.9 \\ -12.5 \\ 355.5 \\ 37.5 \\ -7.5 \end{bmatrix}$$

Since $u_1 = 0$ $F = 0$

free DOF = 4, 5, 6, 9

$$\therefore K_{44} = F - F_{tot}$$

$$K_{44} \begin{bmatrix} u_2 \\ v_2 \\ \theta_2 \\ \theta_3 \end{bmatrix} = \begin{bmatrix} -475.5 \\ 321.9 \\ -12.5 \\ -7.5 \end{bmatrix}$$

$$\therefore \begin{bmatrix} u_2 \\ v_2 \\ \theta_2 \\ \theta_3 \end{bmatrix} = \begin{bmatrix} 0.003054 \\ -0.0036 \\ 0.02534 \\ 0.02170 \end{bmatrix}$$

$$\therefore U = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0.003054 \\ -0.0036 \\ 0.02534 \\ 0 \\ 0 \\ 0.02179 \end{bmatrix}$$

Global reaction:

$$K_u + F_{tot} = K \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0.003054 \\ -0.0036 \\ 0.02534 \\ 0 \\ 0 \\ 0.02179 \end{bmatrix} + \begin{bmatrix} -120 \\ -284.4 \\ 200 \\ -475.5 \\ 321.9 \\ -125 \\ 355.5 \\ 37.5 \\ -75 \end{bmatrix}$$

$$= \begin{bmatrix} -6.0613 \\ 341.1428 \\ 23.5554 \\ 475.5 \\ -321.9 \\ 125 \\ -468.54 \\ -109.2428 \\ 75 \end{bmatrix} + \begin{bmatrix} -120 \\ -284.4 \\ 200 \\ -475.5 \\ 321.9 \\ -125 \\ 355.5 \\ 37.5 \\ -75 \end{bmatrix} = \begin{bmatrix} -126.06 \\ 56.74 \\ 223.56 \\ 0 \\ 0 \\ 0 \\ -113.04 \\ 18.26 \\ 0 \end{bmatrix}$$

$$\begin{aligned} R_{1x} &= -126.06 \text{ kN} & R_{3x} &= -113.04 \text{ kN} \\ R_{2y} &= 56.74 \text{ kN} & R_{3y} &= 18.26 \text{ kN} \\ M_1 &= 223.56 \text{ kN-m} \end{aligned}$$

Global equilibrium check:

$$\sum F_x = 24 \times 10 - 113.04 - 126.06 = 0$$

$$\sum F_y = -75 + 56.74 + 18.26 = 0 \quad (\text{OK})$$

$$U_1^L = T_1 U_1^G$$

$$= \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0.003054 \\ -0.0036 \\ 0.02534 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ -0.0036 \\ -0.003054 \\ 0.02534 \end{bmatrix}$$

$$\begin{aligned}
 U_2^1 &= T_2 N_2^G \\
 &= I U_2^G \\
 &= U_2^G \\
 &= \begin{bmatrix} .003054 \\ -.0036 \\ .02534 \\ 0 \\ .02179 \end{bmatrix}
 \end{aligned}$$

element wise reaction:

$$K_{11} U_1 + \cancel{q_{11}^F} + \cancel{q_{11}^E}$$

$$K_{11} U_1 + q_{11}^F + q_{11}^E$$

∴ for element 1,

$$K_{11} U_1 + q_{11}^F + q_{11}^E$$

$$\begin{aligned}
 &= K_{11} \begin{bmatrix} 0 \\ 0 \\ 0 \\ -.0036 \\ -.003054 \\ .02534 \end{bmatrix} + \begin{bmatrix} 0 \\ 120 \\ 200 \\ 0 \\ 120 \\ -200 \end{bmatrix} + \begin{bmatrix} -284.4 \\ 0 \\ 0 \\ 63.0555 \\ 284.4 \\ 0 \end{bmatrix} \\
 &= \begin{bmatrix} 341.43 \\ 6.061 \\ 23.555 \\ -341.43 \\ -6.061 \\ 46.058 \end{bmatrix} + \begin{bmatrix} 0 \\ 120 \\ 200 \\ 0 \\ 120 \\ -200 \end{bmatrix} + \begin{bmatrix} -284.4 \\ 0 \\ 0 \\ 63.0555 \\ 284.4 \\ 0 \end{bmatrix} = \begin{bmatrix} 56.74 \\ 126.06 \\ 223.56 \\ -56.74 \\ 113.04 \\ -153.04 \end{bmatrix}
 \end{aligned}$$

for element 2:

$$K_{12} U_{12} + q_{12}^F + q_{12}^E$$

$$\begin{aligned}
 &= K_{12} \begin{bmatrix} .003054 \\ -.0036 \\ .02534 \\ 0 \\ 0 \\ .02179 \end{bmatrix} + \begin{bmatrix} 37.5 \\ 75 \\ 0 \\ 37.5 \\ -75 \end{bmatrix} + \begin{bmatrix} -355.5 \\ 0 \\ 355.5 \\ 0 \\ 0 \end{bmatrix} \\
 &= \begin{bmatrix} 468.54 \\ 10.24 \\ 78.94 \\ -468.54 \\ -10.24 \\ 75 \end{bmatrix} + \begin{bmatrix} 37.5 \\ 75 \\ 0 \\ 37.5 \\ -75 \end{bmatrix} + \begin{bmatrix} -355.5 \\ 0 \\ 355.5 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 113.04 \\ 56.74 \\ 153.04 \\ -113.04 \\ 18.26 \\ 0 \end{bmatrix}
 \end{aligned}$$

